RescueSwarm: An Indigenously Sourced Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

Technical Design Proposal

Swastik Kumar
Department of Electronics & Communication Engineering
Thapar Institute of Engineering & Technology
Patiala, Punjab, India
skumar6_be24@thapar.edu

***Abstract*—India recorded 5.4 million internal displacements from disasters in 2024, the highest figure in twelve years, and the binding constraint on survivor outcomes is the speed with which casualties are located and reached. Ground and boat-based search is slow and hazardous; helicopter search is fast but scarce and expensive. This proposal describes RescueSwarm, a three-aircraft autonomous unmanned aerial system that surveys a designated area, detects and geolocates survivors, and delivers a standardised relief kit to each, under a single operator and without any reliance on cellular or internet infrastructure. The system is designed around a deliberate constraint: indigenous sourcing wherever it is achievable, both because supply-chain sovereignty is a national priority and because domestic lead times are roughly half those of imports, which directly buys flight-test days inside a schedule-limited programme. The configuration described here reaches approximately 36 % Indian content, and I state which lines were deferred and what that omission costs the system rather than the budget. I present the system architecture, the sizing and error models that fix its design point, the simulation evidence obtained to date, and a phased work plan. I am explicit throughout about which figures are measured, which are modelled, and which remain assumed; no aircraft has yet flown, and the proposal is structured so that a reviewer can distinguish demonstrated capability from projected capability without reading the source material. This document does not contain the funding request. The programme is delivered by four tracks, each carrying its own work package and funding request, generated from a single budget model and reconciled to the programme total in continuous integration (Section VIII).**

***Index Terms*—unmanned aerial systems, multi-robot coordination, search and rescue, coverage path planning, RTK geolocation, edge inference, indigenous manufacturing, disaster response**

I. Introduction and Problem Statement

A. The operational gap

India records among the highest disaster-displacement figures in the world. The Internal Displacement Monitoring Centre reports **5.4 million internal displacements in India during 2024** — the highest in twelve years — of which 2.4 million were triggered by a single monsoon flood season in Assam [8]. The scale of the response problem is therefore not in question; what is in question is how quickly the people displaced by it can be found.

Post-flood search and rescue in India is dominated by a single quantity: the elapsed time between the onset of the event and the moment a survivor is both *located* and *reached*. Nationally deployed capability addresses the second of these far better than the first. Boat teams and ground parties can reach a survivor once told where to go, but they search slowly, at high risk to the crew, and with poor coverage of debris fields and rooftops. Rotary-wing aircraft search quickly but are scarce, costly per flight hour, and are typically committed to evacuation rather than to systematic area survey.

The consequence is that *localisation* is the bottleneck. A search asset that can systematically cover an assigned polygon, detect human figures in cluttered post-flood imagery, and report each detection as a coordinate accurate to a few metres would materially change the tasking of every downstream resource.

B. Why a swarm rather than a single aircraft

A single multirotor with a useful payload and a fifteen-minute endurance covers a disappointing area. Coverage scales with the number of simultaneously operating aircraft, but so does operator workload, and a system requiring one pilot per aircraft does not deploy well in a disaster. The engineering problem is therefore not simply flight, but *coordinated autonomous flight under a single operator*: area decomposition, task allocation, deconfliction, consolidated reporting, and graceful degradation when the command link is lost.

C. Why indigenous sourcing is a design constraint, not a slogan

I treat domestic sourcing as a first-class design constraint for two reasons, only one of which is political.

The first is sovereignty: a disaster-response asset whose critical components have a single foreign supplier is a strategic liability, and national policy correctly prioritises domestic capability.

The second is schedule, and in a programme of this length it dominates. My supplier survey puts typical import lead times at four to six weeks against two to three weeks domestically.

In a programme whose binding constraint is an eight-week flight-test window, that difference is not a procurement detail — it is the difference between two test campaigns and one. Indigenisation, in this programme, buys flying days.

That argument cuts against a decision taken later in this document, and I would rather confront it than let a reviewer find it. This build defers the Indian autopilot and Indian propulsion on cost, and the imported parts replacing them carry the longer lead time this section has just described as decisive. The mitigation is structural rather than rhetorical: every long-lead item is ordered in tranche 1, in month one, before the software work that would normally precede procurement is finished. Cost was traded against schedule margin knowingly, and the schedule risk it creates is carried explicitly in Table V rather than assumed away.

D. Contributions of the proposed work

1) A three-aircraft autonomous survey-and-delivery system operable by a single operator with no network dependency, sized line-by-line against market-verified components.
2) A geolocation error budget that treats survivor coordinate accuracy as the primary system output, and an architecture that meets it using a team-owned RTK base rather than any external correction service.
3) A coordination scheme — static equal-area decomposition with boustrophedon coverage and greedy claim-and-lock task allocation — selected for determinism and verifiability over optimality.
4) An evidence discipline in which every published number regenerates from its model in continuous integration, and in which measured, modelled and assumed quantities are separated by construction.

II. Objectives and Scope

A. Primary objectives

O1 Survey a designated area autonomously with three cooperating aircraft under one operator and one ground control station.

O2 Detect human figures in nadir imagery at a ground sample distance sufficient for reliable classification, with onboard inference and multi-frame temporal fusion.

O3 Report each detection as a geodetic coordinate with a circular error probable within a few metres.

O4 Deliver a standardised relief kit to each confirmed survivor location by controlled free-fall release.

O5 Operate with no cellular, internet or cloud dependency at any point in the mission.

O6 Maximise value-weighted indigenous content across the bill of materials without compromising O1–O5, and state explicitly which Indian lines were deferred and what the substitution costs the system.

B. Explicit non-objectives

I state these because proposals of this kind are frequently over-scoped, and because each exclusion is a deliberate engineering decision rather than an omission.

- **Not a casualty-evacuation platform.** The payload class is a relief kit, not a person, and no growth path to the latter is claimed.
- **Not beyond-visual-line-of-sight certified.** The design operates within a bounded geofence; regulatory expansion is future work.
- **Not a night or all-weather system.** The perception stack is visible-band. Thermal imaging is identified as the highest-value future extension but is outside this programme.
- **Not a survivor-triage system.** The system reports position and delivers a kit; it makes no medical assessment.

III. Related Work and State of the Art

A. Coverage path planning

Area decomposition for multi-robot coverage is mature. Divide Areas for Optimal Multi-Robot Coverage (DARP) produces equal-area, connected, robot-specific regions and is the natural reference method [1]. Classical boustrophedon decomposition [2] remains the standard in-region motion primitive for sensor footprints that are rectangular in the body frame.

I adopt a static equal-area partition with boustrophedon in-region coverage rather than a full DARP implementation. This is a deliberate trade of optimality for determinism: a static partition produces an identical, inspectable plan for identical inputs, which is a precondition for the verification regime described in Section VI-A. DARP is retained as design rationale and as a future-work path where dynamic reallocation becomes necessary.

B. Multi-robot task allocation

The Consensus-Based Bundle Algorithm (CBBA) is the canonical decentralised auction method for multi-agent assignment [3]. Its guarantees depend on a consensus phase over a communication graph. For a three-aircraft system with a reliable star topology to a single ground station, I implement greedy claim-and-lock with a deterministic tie-break, which is materially simpler to verify and whose worst-case behaviour on three agents is well within acceptable bounds.

C. Aerial detection of people

Small-object detection from altitude is the governing difficulty: a supine adult at survey altitude occupies a few tens of pixels. Slicing-aided hyper inference (SAHI) and related tiling strategies substantially improve small-object recall by running detection on overlapping crops rather than a downsampled full frame [4]. Purpose-built aerial person datasets such as SARD [5] and the broader VisDrone benchmark [6] provide the transfer-learning base; neither contains post-flood Indian terrain. A field data campaign is therefore the right answer; it is *deferred* from this build (Section VIII), which is why detection recall remains a modelled quantity and the leading entry in the risk register.

TABLE I
AIR VEHICLE DESIGN POINT (PER AIRCRAFT)

Parameter	**Value**
Configuration	Quadrotor
Rotor diameter	18 in
Maximum take-off mass	6.36 kg
Fleet mass (3 aircraft)	19.08 kg
Regulatory fleet cap	25 kg
Battery	6S3P Li-ion 21700, 292 Wh
Pack nominal voltage	21.6 V (18.0–25.2 V)
Pack mass	1 449 g
Cells per aircraft	18
Peak pack current	115 A (8.5 C) at T/W 2.0
Static thrust-to-weight	2.0
Hover endurance	15.3 min
Design mission duration	7.7 min
Survey altitude	60 m AGL
Ground sample distance	$\leq$2.0 cm/px
Survey groundspeed	8 m/s
Payload	4 kits $\times$ 200 g
Geofence radius	600 m

D. Airborne payload delivery

The dominant error term in airborne delivery is release state — velocity and wind — rather than release altitude. Mathisen *et al.* demonstrate autonomous ballistic airdrop from a small fixed-wing UAV and report a mean delivery error of 5.5 m, with the release state computed dynamically against wind and vehicle velocity and re-optimised in flight precisely because that term dominates [7]. This is the decisive argument for a multirotor in this role: a rotorcraft can null its groundspeed to near zero before release and thereby remove the dominant error term outright, which a fixed-wing cannot. The delivery budget in Section IV-I exploits exactly this.

E. Positioning

Real-time kinematic correction from a locally surveyed base is standard practice and is the only route to sub-metre relative positioning without an external network [11]. Since the system may not use internet or cellular services, a team-owned base station is not merely preferable but required.

IV. Proposed System Architecture

A. Design point

Table I states the aircraft design point. Every value is produced by a sizing model held in the repository and regenerated in continuous integration; none is transcribed by hand.

This is the reference design point, and one line of it is not yet verified for the aircraft actually being funded. The adopted configuration (Section VIII) uses generic motors that publish no thrust curve, so the thrust-to-weight figure below is a requirement rather than a measurement until the P5 thrust stand runs. Nothing else in the table moves: mass, energy, altitude, speed and payload are all set by the mission.

The thrust-to-weight ratio of 2.0 and the hover endurance of 15.3 minutes against a 7.7-minute design mission are reserve policies rather than performance targets, because the failure mode that ends a competition or a deployment is an aircraft that does not return. The energy reserve is stated precisely in Section IV-B; it, and not the design mission, is what sizes the pack.

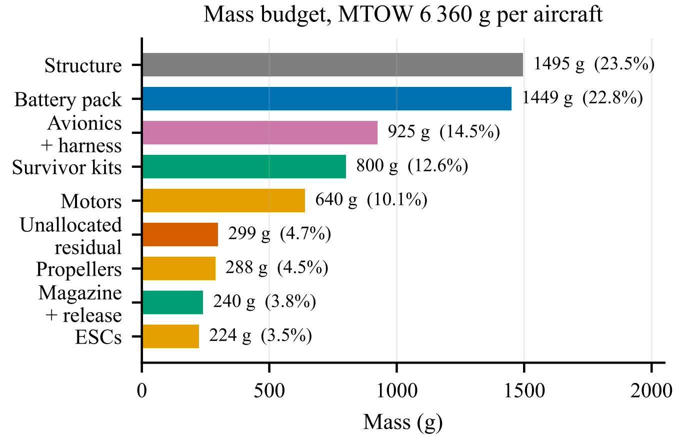

Fig. 1. Mass budget per aircraft. Generated from `docs/sizing/model-output.txt`. The fleet figure of 19 081 g carries 4 919 g of growth allowance against a 24 kg internal target — 26 % build overweight tolerated — inside the 25 kg regulatory cap.

Fig. 1 gives the mass statement behind those figures. Two features drive the rest of the design. Structure and battery together are 46 % of maximum take-off mass, which is why airframe mass fraction and pack chemistry are the two sensitivities carried in the risk register rather than, say, avionics choice.

The itemisation does not close, and the figure shows the gap rather than hiding it. The model's listed items sum to 6 061 g against a 6 360 g maximum take-off mass, leaving 299 g — 4.7 % — unallocated. That residual is small enough not to threaten the fleet margin and large enough that it should be attributed rather than absorbed silently; closing it is a P1 action. The survivor kits are 800 g — 12.6 % of MTOW and fixed by the mission at four kits of exactly 200 g — so payload is an input to the sizing loop, not an output of it. This is why propulsion and energy resist cost reduction in a way that avionics does not: they are sized by a payload that is not negotiable, so cheaper parts of the same class are available but a smaller aircraft is not.

B. Energy: the decisions behind the pack

Section V derives the pack requirement in full — the reserve policy that sizes it, the nameplate energy floor, and the continuous current figure that disqualifies most candidate packs. The re-sweep term in that policy exists because a search finding nothing on its first pass is a normal outcome rather than a failure; the loiter term exists because a recovery queue of three aircraft is not instantaneous.

Two things about the result are decisions rather than arithmetic, and they belong here rather than in a derivation.

Why cylindrical Li-ion rather than LiPo. I record this as a trade rather than a preference, because LiPo wins part

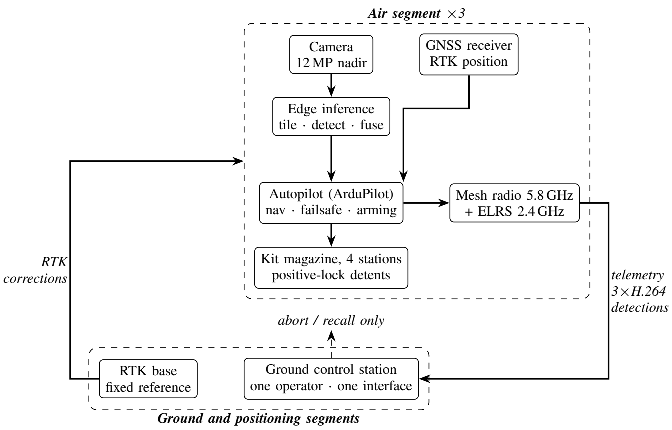

Fig. 2. System architecture. The only permitted downlink commands are abort and recall; the ground station is architecturally incapable of originating a retask, waypoint change or drop command. Detection runs onboard, so the downlink serves the operator rather than the algorithm.

of it. A LiPo pack of equal energy would roughly halve the internal resistance — about 2.5 V of sag at peak against 4.6 V — and would carry far more current headroom, and it would remove the pack-assembly step entirely. It loses on three grounds that matter more to this programme: it is approximately 400 g heavier per aircraft at equal energy, which spends fleet margin; its cycle life is two to three times worse across a long flight-test campaign; and it is not domestically sourced, which would remove one of only four remaining Indian lines in Section VIII. The decision is therefore made on campaign cost and sourcing, not on the electrical argument, and I state it that way because the electrical argument points the other way.

One correction, recorded rather than absorbed. The reserve policy was previously reported at 159 Wh. That figure was computed at the sizing solve, before the pack was rounded up to a buildable 18 cells and maximum take-off mass settled roughly 0.7 kg heavier; the model checked a constraint against a design point it then abandoned. Re-evaluated at the adopted mass the requirement rises by roughly a fifth, to the figure given in Section V, and the chemistry-agnostic nameplate floor rises with it. The pack still passes, on a materially smaller margin than the earlier figure implied. The model now performs this check at the final design point, and a pack bought against the superseded number would have missed the policy.

C. Segment architecture

The system comprises three segments, shown in Fig. 2.

1) Air segment: Three identical quadrotors, each carrying a Pixhawk-standard autopilot running ArduPilot [10], a dual-band NavIC/GNSS receiver providing RTK position, a nadir-mounted 12 MP camera, an edge inference module, a four-station kit magazine with positive-lock detents, a 5.8 GHz mesh radio for telemetry and video, and a 2.4 GHz ExpressLRS link carrying safety-pilot control and abort independently of the autonomy stack.

TABLE II
GEOLOCATION ERROR BY CONFIGURATION

Case	Configuration	RSS error
A	No RTK, single-frame	3.09 m
B	RTK, single-frame	∼1.3 m
C	RTK + multi-frame fusion + calibrated ground plane	**0.91 m**

Case C assumes moving-baseline heading. That receiver is deferred (Section VIII), which substitutes a magnetometer-class attitude term and gives **0.94 m** — an 0.03 m penalty, and the reason the deferral is judged affordable.

2) Ground segment: A single ground control station presenting one unified operator interface. The station is architecturally incapable of originating a retask, waypoint change or drop command; abort and recall are the only permitted downlink commands, and are implemented as an explicit exception. This is a structural safety property rather than a policy: the relevant code paths are absent from the deployed build, not merely guarded.

3) Positioning segment: A team-owned RTK base station, positioned and set to a fixed reference from its first three-dimensional fix within the setup window. Launch is gated on a three-dimensional fix rather than an RTK fix; the first geotag is gated on RTK-fixed, and detections made before convergence are geotagged in float and re-fused once fixed. This decoupling recovers approximately ninety seconds of setup time that a survey-in procedure would otherwise consume.

D. Geolocation error budget

Survivor coordinate accuracy is the primary system output, and the error budget is therefore the governing analysis of the whole design. Total error is the root sum of squares of GNSS position error, boresight and lever-arm residual, attitude error, terrain-model error and timing skew.

Table II shows the dominant conclusion: positioning quality, not optics, governs. This is what makes RTK a scoring requirement rather than an optimisation, and it is why the RTK receiver and its base station survive every cost-reduction pass while a great deal else does not. The *second* receiver, which supplies moving-baseline heading, is a different case and is deferred to a measurement — see Section VIII.

Multi-frame fusion is effectively free. At 60 m altitude and 8 m/s groundspeed, forward overlap yields on the order of fourteen independent observations of each target per pass even at a conservative 2 Hz inference rate. Those observations are used both for temporal confirmation, which suppresses false positives, and for geolocation averaging.

Fig. 3 shows both effects quantitatively. Panel (a) makes the positioning argument unarguable: losing the RTK fix moves the error by nearly an order of magnitude, and a 3-D-only fix at 3.67 m RSS consumes most of the 5 m delivery allowance before ballistics or position-hold error are counted. Panel (b) shows why fusion is worth having but not worth over-investing in — it divides the random terms by $\sqrt{n}$ and leaves boresight and target extent untouched, so the curve departs from the ideal $1/\sqrt{n}$ line and flattens by about five frames. Twenty frames is not twice as good as five, and the roughly fourteen frames the geometry supplies for free sit comfortably past the knee.

E. Perception and inference budget

The sensor is a 12.3 MP module, 4056×3040 on a **1.55 µm** pitch behind a 6 mm lens. Running detection on the full-resolution frame would require 48 tiles and is far beyond the thermal and compute envelope of any module in this mass class, so the design downsamples by two before tiling. That cuts the tile count to 12 per frame at 20 % overlap; at a 2 Hz inference rate this is 24 inferences per second at 640×640, within the demonstrated throughput of the accelerator selected for this build. Tile count depends on pixel count alone, so it does not move with altitude.

What the detector is handed does move with altitude, and the effect is larger than it looks:

AGL	GSD (cm/px)	Target (px)	Area (px^2)	COCO class
40 m, 2×	2.07	82	1 990	medium
60 m, 2×	3.10	55	884	**small**

The size class, not the pixel count, is the number that matters. Published recall figures report average precision *by* size band, and small-object AP is routinely far below medium on the same model. At 60 m the target falls below COCO's 32^2 px^2 small-object threshold; at 40 m it sits comfortably in the medium band. Any recall figure this programme benchmarks against is only comparable if the size band matches.

A correction is recorded here rather than absorbed. Earlier revisions quoted 2 cm/px and 85 pixels at 60 m, falling to 47 pixels after downsampling. Those figures were computed for a 1.82 µm sensor assumed in the sizing chapter — the same pixel *count* at a different pixel *size* — and not for the sensor the bill of materials buys. The corrected figures are above. The substitution cuts both ways: a smaller pixel gives finer ground sampling and more pixels on target, but a narrower field, so coverage costs more transects. Neither effect was chosen; both follow from the part.

This is the single most benchmark-sensitive figure in the design. Published throughput for comparable models on comparable hardware spans roughly 24 to 41 frames per second, and the lower end of that range leaves no margin. Hardware benchmarking before design freeze is accordingly a funded, scheduled activity rather than an assumption.

F. Why the compute stack is three parts

The air vehicle carries a sensor, a single-board host and a neural accelerator. On an aircraft this size that invites the question of why it is not one part, and the answer is that each does something the others cannot.

The accelerator is an **M.2 module whose only interface is PCIe**. It has no camera input, no image signal processor and no general-purpose cores; it executes a compiled graph on tensors a host hands it. The sensor emits a raw Bayer-mosaic stream, which is not an image until something demosaics, white balances and scales it. And the tiling that the whole detection approach rests on — cutting each frame into twelve overlapping crops, submitting them, and reassembling detections back into frame coordinates — is host work by construction, as is every other thing the aircraft does: the phase machine, MAVLink routing, the mesh, survivor claim and release, the delivery excursion and the telemetry the ground station consumes.

So the three parts are a sensor, an image pipeline with a CPU, and a matrix engine. Removing any one of them does not simplify the aircraft; it stops the detection chain working.

One alternative would genuinely remove a box, and it was rejected on a measurable ground rather than a preference. Sensors now exist with an inference engine on the die — the imager runs the network itself and emits detections instead of pixels, at roughly 40 % lower power than a discrete accelerator, and it would delete the accelerator line entirely. It does not fit this mission: on-sensor engines carry a model of a few megabytes and perform a *single* inference per capture at a fixed input size. This design needs **twelve** inferences per frame, because a survivor occupies a few tens of pixels in a 12 MP frame and tiled inference is the only thing that makes that target detectable at all (Section IV-G). An imager that cannot tile its own output is being asked to do the one thing the architecture exists to avoid. It also removes nothing else: the host is still required for everything in the paragraph above.

That trade is worth stating because it is the obvious economy to propose, and the reason it fails is specific to small-object search rather than general.

G. Onboard processing pipeline

Fig. 4 shows what runs where. The division of labour is the part worth stating plainly, because it is what keeps the search phase deterministic: **the autopilot owns the flight, and the companion computer never touches it during the sweep.** Transects are uploaded as an ordinary AUTO mission during setup, and the aircraft would fly the entire search pattern with the companion powered off. The companion observes, detects, and takes control only for a delivery excursion — a bounded GUIDED interruption that returns the aircraft to the same AUTO mission afterwards.

That boundary matters for two reasons. A companion crash or thermal throttle costs detections, not control. And the search path is reproducible from the mission file alone, which is what makes the coverage and deconfliction results in Section VI-A checkable rather than emergent.

Detection does not gate flight. Inference runs on captured frames while the aircraft continues its transect. Losing frames to a thermal event or a slow model costs looks at a target, and the fusion requirement of twelve looks per pass carries enough redundancy to absorb some of that; it does not stall the sweep or extend the mission.

The accelerator is a substitution the inference budget has not yet been re-run against. The tiling arithmetic above

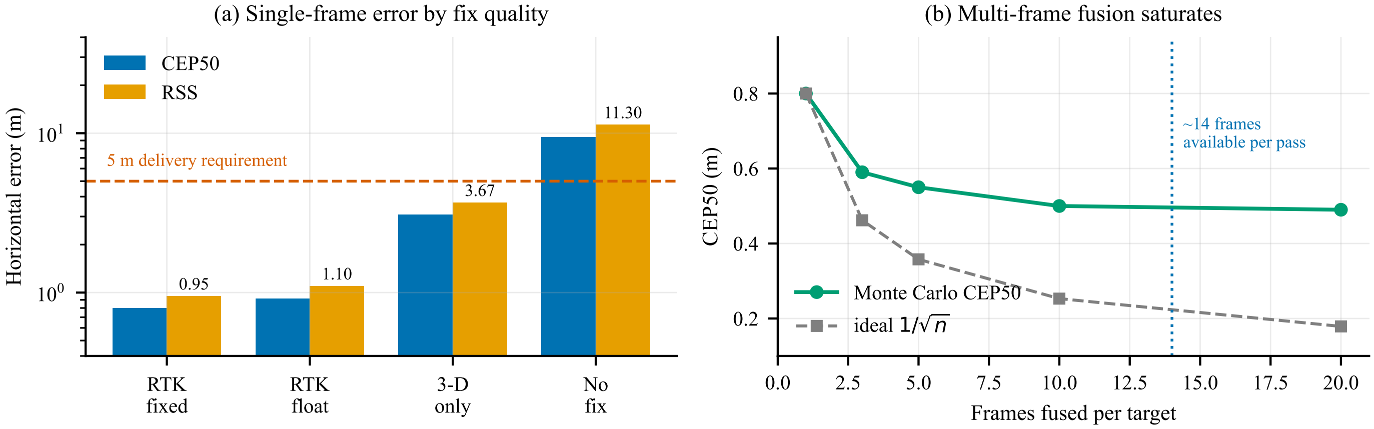

Fig. 3. Geolocation error, from a Monte Carlo run through the same projection code the aircraft flies. (a) Single-frame horizontal error against GNSS fix quality, typical attitude, on a logarithmic scale; the dashed line is the 5 m delivery requirement. (b) Effect of fusing multiple observations of the same target, RTK fixed. Generated by `figures/make_figures.py` from `docs/sizing/geotag-accuracy-output.txt`.

was derived against a Jetson-class module, whose published throughput for comparable models spans roughly 24 to 41 frames per second. The adopted configuration instead pairs a 26-TOPS accelerator with a single-board host, which is a different architecture with a different throughput curve and a lower sustained power draw. The 24-inferences-per-second requirement is therefore stated against hardware that has not been benchmarked, and P5 benchmarking on the parts actually purchased is the gate before design freeze. Table V carries this rather than the text implying the figure is settled.

H. Communications

Regulatory rules require the ground station to display a live camera feed from *each* aircraft. Three independent MJPEG streams would demand 4.5–6 Mbps against a link budget of approximately 2.5 Mbps. The system therefore uses H.264 encoding served through a media gateway, carrying three concurrent WebRTC streams in approximately 2.7 Mbps, with per-feed rate reduction under congestion rather than feed switching. Detection runs onboard; the downlink serves the operator, not the algorithm.

Band diversity is provided without a third radio. The 5.8 GHz mesh carries video, telemetry and coordination; a 2.4 GHz ExpressLRS link carries safety-pilot control and abort, and is independent of the autonomy stack. Both operate in delicensed bands and both comfortably cover the 600 m geofence.

The abort path is not an assumption about the link; it is a documented mode of it. ExpressLRS carries MAVLink natively from version 3.5, so the same receiver that gives the safety pilot manual control also carries the autopilot's command and telemetry stream — the ArduPilot side is a serial port set to MAVLink v2, nothing more [9]. The earlier AirPort mode achieves the same thing as a transparent serial bridge and is no longer the recommended route for MAVLink. Naming the mechanism matters here: deferring the sub-gigahertz radio (Section VIII) rests entirely on this link being able to do both jobs. A further link in the 865–867 MHz short-range-device band would add a third independent path and is the correct extension when funding allows; it is deferred rather than fitted, and Section VIII records the reasoning.

I. Payload delivery

Kits are released at 6 m above ground level, gated on a residual groundspeed below 0.3 m/s. At that release state, ballistic dispersion contributes approximately 0.32 m to total delivery error; combined with geolocation and position-hold error the root-sum-square is approximately 1.5 m with RTK active. Release altitude is chosen low enough that wind drift remains under a metre in realistic conditions, and high enough for safe clearance above a person and above debris.

Positive-lock mechanical detents on each station ensure that a power interruption cannot release a kit. This is treated as a safety-critical item and is excluded from cost reduction by policy.

J. Mission sequence

Fig. 5 shows the autonomous mission sequence. Two properties are worth drawing out. First, the launch stagger lives in the mission file as an explicit delay before each take-off command, not in operator timing — this is what converts a 1.31 m closest approach into the 64.80 m of Fig. 6. Second, every terminal branch converges on return-to-launch: link loss, low battery and operator abort share one recovery path rather than three. Third, the battery branch is *staged* rather than single-valued: the reserve threshold commands return-to-launch, and a second, lower threshold commands an immediate landing where the aircraft stands. A pack that cannot reach the pad should land under control at the point it admits this, rather than descend uncommanded partway home.

V. Sizing and Analysis

This section gives the derivations behind the design point rather than only its result. Every figure regenerates from `tools/sizing-model/rescueswarm_sizing_model.py`;

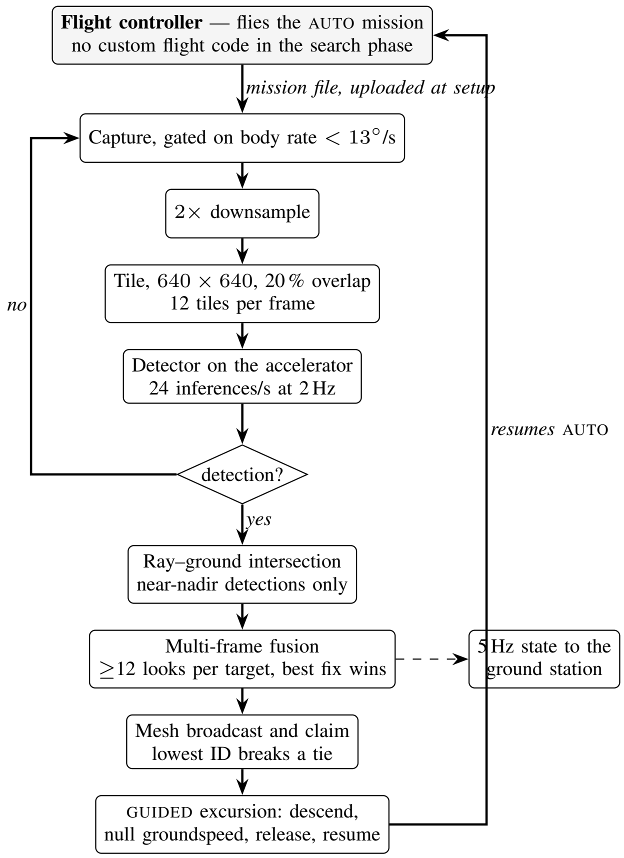

Fig. 4. Onboard processing. The autopilot flies the search pattern unaided; the companion computer detects, geolocates and claims survivors, and interrupts the mission only to deliver. Detection runs alongside the sweep and never gates it — a missed frame costs a look, not a waypoint. The capture gate suppresses turn arcs, where the ground-plane assumption is least valid and angular smear is worst. One pixel subtends 13.63 mdeg on this sensor, so at $1/1000$ s a body rate of 13.6°/s is exactly one pixel of angular blur; the gate is set at 13°/s to sit under it.

the section itself is generated, so a constant cannot change in the model without changing here.

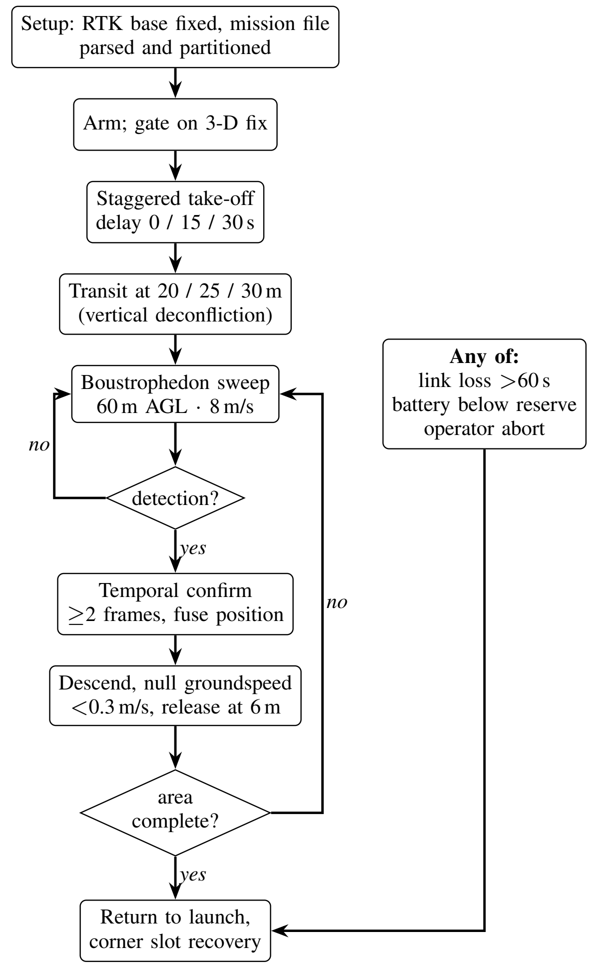

Fig. 5. Autonomous mission sequence. The take-off stagger is encoded in the mission file rather than in operator timing, and all three abnormal conditions converge on a single return-to-launch path.

A. *Constants and where they come from*

Symbol	**Value**	**Basis**
ρ	1.150 kg/m^3	∼30 °C at 300 m density altitude; an Indian summer field, not sea level
FM	0.60	Rotor figure of merit, mid-range of the 0.5–0.7 literature band
η_{mot}	0.82	BLDC efficiency near the hover operating point
η_{esc}	0.95	Electronic speed controller
P_{avio}	55 W	Companion computer, autopilot, GNSS, camera, radios, servos
DoD	80%	Usable depth of discharge; land at 20 % state of charge
T/W	2.0	Static thrust-to-weight, a reserve policy

Two of these are deliberately unflattering. Air density is taken at a hot-day altitude rather than sea level, which *increases* the power required; and the figure of merit is mid-band rather than optimistic. Sizing errors that favour the design are the ones that are discovered on the flight line.

B. Hover power, from momentum theory

The rotor disk area for 4 rotors of diameter 18 in ($D = 0.4572\,\mathrm{m}$) is

$$A = N\pi\left(\frac{D}{2}\right)^2 = 0.6567\ \mathrm{m}^2, \qquad \text{disk loading} = \frac{m}{A} = 9.69\ \mathrm{kg/m}^2.$$

Momentum theory gives the induced power to generate thrust T across that disk. Dividing by the figure of merit converts ideal induced power to shaft power:

$$P_{\text{shaft}} = \frac{T^{3/2}}{\mathrm{FM}\sqrt{2\rho A}} = \frac{(62.4)^{3/2}}{0.60\sqrt{2(1.150)(0.6567)}} = 668\ \mathrm{W},$$

where $T = mg = 62.4\,\mathrm{N}$ at the 6.36 kg maximum take-off mass. Adding the drivetrain and the avionics load,

$$P_{\text{elec}} = \frac{P_{\text{shaft}}}{\eta_{\text{mot}}\eta_{\text{esc}}} + P_{\text{avio}} = \frac{668}{0.779} + 55 = 913\ \mathrm{W},$$

which at the 21.6 V pack nominal is a hover current of $I = P/V = 42.2\,\mathrm{A}$. Power loading is 7.0 g/W.

C. Peak power and the current the pack must supply

Induced power scales as $T^{3/2}$, so at the design thrust-to-weight of 2.0 the shaft power rises by $2.0^{3/2}$:

$$P_{\text{peak}} = \frac{(2.0\,T)^{3/2}}{\mathrm{FM}\sqrt{2\rho A}\,\eta} + P_{\text{avio}} = 2481\ \mathrm{W} \implies I_{\text{peak}} = 114.8\ \mathrm{A}.$$

That current, not the energy, is what disqualifies most candidate packs. It is a continuous requirement rather than a burst rating: the aircraft must be able to hold 2.0 × weight, not pulse it. The adopted pack supplies 135 A, a margin of 18 %, and the per-motor share sets the controller rating at 29 A peak.

This is also where a smaller pack fails. At 13.5 Ah the peak is 8.51 C and hover is 3.13 C; a pack half the size would need twice the C-rate to deliver the same amperes. Capacity is how the design buys down discharge rate.

D. Propulsion

$$T_{\text{total}} = (T/W)\,mg = 125\,\mathrm{N} = 12.72\ \mathrm{kgf}, \qquad T_{\text{motor}} = 3.18\ \mathrm{kgf}.$$

Hover demands 1.59 kgf per motor, which is **50 % of maximum** — inside the 45–55 % band where propeller efficiency and thermal margin are both reasonable. Sitting far below that band means an oversized and heavy propulsion system; far above it means no authority left for attitude control in gusts.

E. The coupled mass and energy solve

Pack mass, take-off mass and mission energy are mutually dependent: a heavier pack raises hover power, which raises the energy required, which raises pack mass. The model iterates to a fixed point rather than assuming one.

The reserve policy is what closes it. The pack must carry the nominal mission, one complete re-sweep of the search area, and four minutes of loiter, inside the usable window:

$$\underbrace{105.5}_{\text{nominal}} + \underbrace{22.1}_{\text{re-sweep}} + \underbrace{60.8}_{\text{loiter}} = 188.4\ \text{Wh usable,}$$

against 233.3 Wh available at 80% depth of discharge — a margin of **24 %**. Expressed independently of chemistry, any candidate pack must therefore provide **236 Wh** nameplate at **6S** and **114.8 A continuous**.

The adopted implementation is 18 cells in 6S3P: 292 Wh, 13.5 Ah, 1449 g, 21.6 V nominal (18–25.2 V). Hover endurance is 15.3 min against a 7.7 min design mission consuming 105.5 Wh.

At 6.36 kg per aircraft the fleet is **19.08 kg** against the 25.0 kg regulatory cap, leaving 24 % margin.

F. Camera and optics

Everything optical follows from three numbers: pixel count, pixel pitch and focal length. Active area is count times pitch,

$$w = 4056 \times 1.55\,\mu\mathrm{m} = 6.287\ \mathrm{mm}, \quad h = 4.712\ \mathrm{mm}, \quad d = 7.86\ \mathrm{mm},$$

and the fields of view are exact arctangents, with no small-angle approximation:

$$\theta = 2\arctan\left(\frac{s}{2f}\right) \Rightarrow \mathrm{HFOV} = 55.3^\circ,\ \mathrm{VFOV} = 42.9^\circ,\ \mathrm{DFOV} = 66.4^\circ.$$

Sensor and ground are similar triangles about the lens, so ground sample distance at altitude H is

$$\mathrm{GSD} = \frac{p\,H}{f} = \frac{1.55\,\mu\mathrm{m} \times 40\,\mathrm{m}}{6\,\mathrm{mm}} = 1.03\ \mathrm{cm/px},$$

giving a $41.9 \times 31.4\,\mathrm{m}$ footprint and a 1.7 m supine adult subtending **165 px** at 40 m. One pixel subtends 14.80 mdeg.

Focus is not a design problem, and the arithmetic says so. The hyperfocal distance,

$$H_{\text{hyp}} = \frac{f^2}{Nc} + f = 4.15\ \mathrm{m},$$

taking the circle of confusion as two pixels rather than a film-era constant, puts everything from 2.08 m to infinity in acceptable focus. The aircraft never operates below 30 m. A fixed lens set once is correct at every altitude flown, and a focus mechanism would add a moving part, a power draw and a failure mode that cannot be detected from the air.

Motion blur does not bind either. Holding smear under one pixel requires $t_{\text{exp}} \le \mathrm{GSD}/v$, which at 8 m/s is 1.29 ms — a shutter of $1/774$ s. The requirement already mandates $1/1000$ s or faster.

Rolling shutter is small but not zero. Reading the frame out over 25 ms while the aircraft moves gives 0.20 m of top-to-bottom skew, about 19 px. Negligible for detection; *not* negligible for geolocation, where a detection's row position carries an along-track bias unless the pipeline corrects for readout row.

Temporal sampling is where the budget does not close. A target is in frame for the along-track footprint divided by ground speed, so at 40 m and 2 Hz it receives **7.9 looks** against the 12 that multi-frame fusion requires. Meeting that count needs

$$r \geq \frac{n\,v}{D} = 3.06\ \text{Hz},$$

and the trade — higher capture rate, lower ground speed, higher altitude, or a relaxed look count — is open and recorded as such.

G. Payload release ballistics

A kit of 200 g tumbling with drag coefficient 1.05 and reference area 117 cm^2 has ballistic coefficient $\beta = m/(C_d A) = 16.3\,\text{kg/m}^2$ and terminal velocity

$$v_{\text{term}} = \sqrt{\frac{2mg}{\rho C_d A}} = 16.2\ \text{m/s}.$$

Integrating the fall with quadratic drag from a 6 m release gives 1.15 s to impact at 9.7 m/s, with a crosswind drift of 3.24 m at 3 m/s and 6.13 m at 6 m/s.

The governing sensitivity is release velocity, not release height. A residual ground speed of 0.5 m/s displaces the impact point by about 0.53 m, whereas a full metre of altitude error contributes under 4 cm. This is the whole argument for hover-and-drop: a multirotor can null its ground speed before release and a fixed-wing cannot.

H. Structure, geometry and control authority

Tip clearance of 30 mm between adjacent 18 in rotors sets a diagonal wheelbase of 689 mm and an arm length of 0.345 m. At the design thrust-to-weight the root bending moment is

$$\mathcal{M} = T_{\text{motor}}\,L = 10.7\ \text{N m}.$$

For a 25×23 mm carbon tube, $I = \pi(D_o^4 - D_i^4)/64 = 5438\,\text{mm}^4$, so the bending stress is $\sigma = \mathcal{M}c/I = 25\,\text{MPa}$ against roughly 600 MPa for unidirectional carbon — a safety factor of about **24**.

Bending is therefore not the structural driver, and the design should not be optimised as though it were. Joint and clamp design, and vibration fatigue at the arm root, are what actually govern; a tube chosen to satisfy the bending case by a factor of 24 is being chosen for stiffness and handling, not strength.

Releasing one kit from a magazine offset from the centreline shifts the centre of gravity by under 2 mm, requiring a few tens of grams of differential thrust to trim — trivial in magnitude, but a *step* change that excites the attitude loop. The mitigation is to mount the magazine on the centre-of-gravity axis and hold position briefly after release rather than to trim harder.

I. Radio link budget

Free-space path loss at the 600 m design slant range is $\text{FSPL} = 20\log_{10} d + 20\log_{10} f + 32.44$:

Link	FSPL	Role
5.8 GHz mesh	103.3 dB	Data and video; margin is thin and degrades first
2.4 GHz mesh	95.6 dB	Fallback
868 MHz safety	86.8 dB	Command of last resort; large margin by design

The ordering is the design intent: the highest-rate link is the one permitted to fail, and the lowest-rate link is the one that must not. Video degrades before telemetry, and telemetry before command.

J. What this section does not establish

Every figure above is *modelled*. The thrust figures assume a motor that publishes no thrust curve, and the detection figures assume a recall that has not been measured. The sizing is internally consistent and its arithmetic is checkable; that is a different claim from saying the aircraft has been shown to do these things, and this document does not make the second claim anywhere.

VI. Methodology and Work Plan

The programme is organised into ten phases. Phases P1–P5 are analysis, procurement and integration; P6–P10 are progressively more demanding flight test. The eight-week flight-test window spanned by P6–P10 is the binding schedule constraint on the entire programme, and the work plan is arranged to protect it.

A. Verification strategy

The programme's verification discipline is, I believe, its most transferable contribution, and it exists because of a specific and recurring class of defect.

On four separate occasions during preparatory work, a configuration artefact was found to be correct, unit-tested, reviewed, and *not in the execution path* of the running system. In one instance a validated failsafe parameter set was never loaded by any simulation, so every test ran against stock defaults in which the low-battery action was disabled. In another, a message-routing configuration started cleanly and forwarded zero messages for weeks. Each of these reviewed clean and had passing tests around it.

The countermeasure adopted is procedural rather than technical: no capability is considered demonstrated until the real artefact has been exercised end to end and the resulting values read back off the running system. Consequently every claim in this proposal carries an explicit evidence class, and those classes are maintained mechanically rather than editorially:

- **Measured** — observed on a running system, in software or in hardware-in-the-loop simulation.

TABLE III
PROGRAMME PHASES

Ph.	Title	Principal exit criterion
P1	Requirements and sizing	Design point frozen; all models regenerate
P2	Ground segment	Single-operator interface demonstrated against three simulated aircraft
P3	Autonomy	Deterministic plan generation; deconfliction verified in simulation
P4	Fault handling	Link-loss and low-battery behaviour verified by fault injection
P5	Procurement and build	Three airframes assembled and weighed; thrust stand complete
P6	First flight	Hover, trim, vibration and thermal characterisation
P7	Perception field trials	Detection recall measured on real imagery against fabricated targets on RTK-surveyed ground
P8	Delivery trials	Delivery CEP measured against a surveyed target mat
P9	Full mission rehearsal	End-to-end mission with all three aircraft, no operator intervention
P10	Setup and contingency	Setup drill timed over repeated runs; margin recovered

- **Modelled** — computed by a model whose inputs are themselves assumptions, and which regenerates in continuous integration.
- **Assumed** — neither measured nor modelled; carried explicitly as a risk.

A continuous-integration job enforces the rule that every published number regenerates byte-for-byte from its model, so that changing a model without committing its regenerated output fails the build. Thirteen model outputs are currently under this gate.

The same discipline now covers the cost basis. Every figure in this section is imported from a single module, which also generates the cost workbook and the budget charts; nothing is restated in two places. That was not true earlier in this programme — the workbook and this document drifted until they disagreed about the headline figure — and the fix was structural rather than editorial, because an inconsistency that can be reintroduced by hand eventually is.

VII. Preliminary Results

Substantial work has been completed before this request. I summarise it here with the evidence class of each result stated, because a claim whose basis is declared can be independently checked, whereas a claim whose basis is left implicit cannot be.

A. Measured in simulation

- **Low-battery failsafe returns the aircraft to the pad.** Verified across three simultaneous software-in-the-loop autopilots; return-to-launch triggered at 10 809 mAh of a 10 800 mAh planned consumption.
- **Three aircraft on one ground station.** A single telemetry router carried three distinct system identifiers to one ground station port with all three arriving.
- **Inter-aircraft separation at launch.** Staggered take-off delays encoded in the mission file, rather than in operator timing, increased the closest airborne approach during launch from 1.31 m to **64.80 m** (Fig. 6). Both figures are recomputed directly from the committed telemetry.
- **Separation during the sweep.** Minimum airborne separation away from the pad is **29.19 m**, with staggered transit altitudes of 20/25/30 m providing vertical deconfliction independent of the lateral plan.
- **Recovery deconfliction.** Assigning landing slots to the corners of the recovery pad rather than in a row increased slot spacing from 1.22 m to 2.61 m at no cost, raising the measured minimum approach from an overlapping 0.83 m to **2.27 m** — a clearance of 1.22 m between airframes (Fig. 7). Separation while stacked in the air over the pad improves by the same mechanism.
- **Ground station.** All eight required operator displays were verified against three simultaneous autopilots, carrying three concurrent H.264 streams within the link budget, with offline map tiles (Fig. 9).

1) Pad containment has exactly zero margin: Fig. 7 shows the corner slots sitting precisely on the inset square. Nominal edge clearance is 0.523 m — one airframe radius — so no part of a stationary aircraft lies outside the box, and the geometric requirement is met by construction.

That is compliance with no margin, and it is worth stating plainly. The rules admit no part of any aircraft outside the box during launch and landing. Touchdown dispersion of ± 0.5 m, the figure this programme uses elsewhere for landing accuracy, would put up to half a metre of airframe over the edge.

The two constraints pull against each other: moving the slots inward to buy edge margin costs inter-aircraft spacing, which is precisely why they were moved to the corners. Resolving it needs a *measured* touchdown dispersion rather than an assumed one, and that is a P6 activity. It is recorded here because the geometry looks safe on the figure and is only safe on average.

B. Modelled

Endurance, hover power, mass budget, thermal margin, link budget, delivery dispersion and geolocation error are all modelled and regenerate from source. The geolocation Monte Carlo reconciles with the independent analytic budget to within one per cent, and the geotag projection has been verified self-consistent between two independent formulations to 7.8×10^{-10} m.

C. Assumed, or not yet implemented

Boresight calibration residual, achieved RTK accuracy in the field, and battery internal resistance under Indian ambient conditions are all assumed, and each is a funded activity in P5–P8. **Detection recall on real post-flood imagery is also**

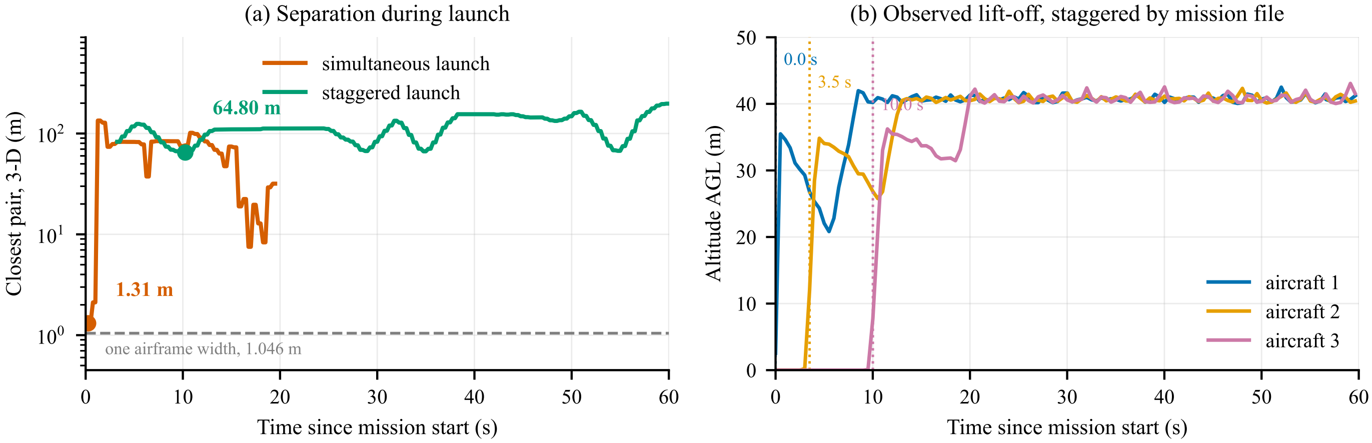

Fig. 6. Launch deconfliction, recomputed from the committed telemetry of three ArduCopter software-in-the-loop autopilots. (a) Closest pair in three dimensions, airborne aircraft only, on a logarithmic scale. Launching simultaneously brings two aircraft within 1.31 m — inside one airframe width, marked — while staggering take-off in the mission file holds the closest pair to 64.80 m. (b) The mechanism: lift-off times observed in the same recording. The mission file commands delays of 0/15/30 s; the aircraft are observed leaving the pad at 0.0/3.5/10.0 s, and **the figure plots what was observed rather than what was commanded** — see Section VII-C.

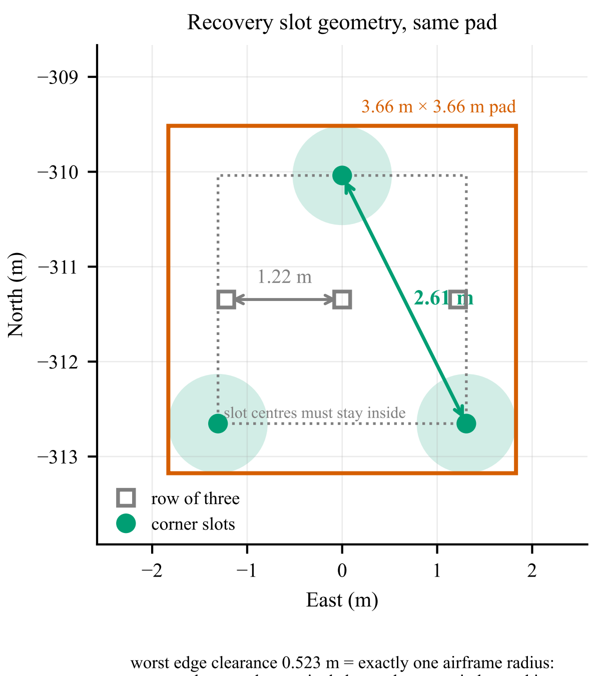

Fig. 7. Recovery slot geometry, drawn from the two committed recordings. Slot centres must stay half an airframe inside the pad edge, so they live in a 2.61 m square: a row across that square gives 1.22 m of spacing, its corners give the full 2.61 m. Shaded discs are the 1.046 m airframe footprint at the corner slots. The measured closest approach at any time rises from an overlapping 0.83 m to 2.27 m — twice the separation, on the same pad, at no cost.

assumed, and its field campaign is deferred rather than funded (Section VIII); it is therefore the one assumption this programme does not currently plan to retire, and it leads the risk register accordingly.

One ground-segment function is explicitly *not* implemented and is visible as such in Fig. 9: the abort and recall controls record operator intent in the mission log but transmit nothing, because no downlink command path is yet wired. Recovery today depends on the safety pilot's transmitter. Closing this is a P4 activity, and it now runs over the 2.4 GHz ExpressLRS link rather than the deferred sub-gigahertz radio — which is what makes that link load-bearing for safety and not merely a control convenience.

D. The honest summary

Nothing in this programme has flown. Every "measured" result above was measured in simulation or in software. That rules out whole classes of logical and integration error and rules out none of the physical ones. The build phase exists principally to convert the modelled and assumed rows of this section into measured ones.

VIII. Scope of This Build

This document describes an integrated system. **The funding request is not part of it.** The programme is delivered by four tracks, each of which carries its own work package and its own funding request, generated from a single budget model rather than restated by hand: `docs/proposal/tracks/`. The four track asks reconcile to the programme total exactly, and that reconciliation is asserted in continuous integration.

What belongs here is not what the build costs but *what it does and does not contain*, because several arguments earlier in this document depend on it.

A. What this build deliberately does not include

Each of the following was deferred rather than judged unnecessary. Stating them is not padding: a reviewer should be able to see what the lowest-cost configuration gives up, and in two cases the omission changes what this document is entitled to claim.

The field-data campaign and the ground-truth apparatus are both deferred, and the consequence is stated plainly: ***detection***

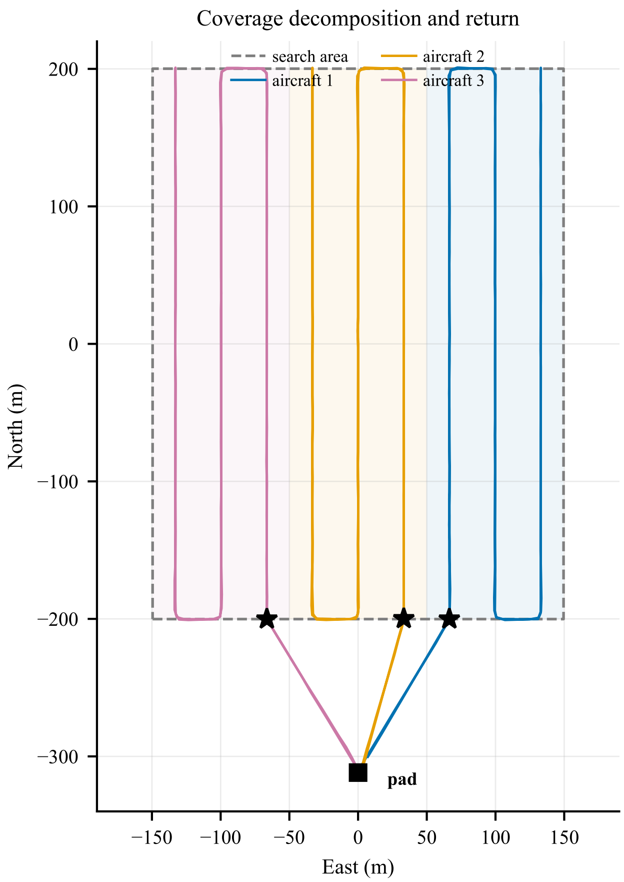

Fig. 8. Coverage decomposition and return geometry, drawn from the committed telemetry. Each aircraft is assigned an equal-area strip (shaded) and flies a boustrophedon pattern within it; each strip is swept twice, on opposing headings, so that boresight error cancels rather than accumulating. Stars mark where each sweep ends. Enumerating the four start-side and direction combinations and keeping the one that finishes nearest home brings all three aircraft to within 117–130 m of the pad, against 516 m and 540 m for two of three under the previous index-keyed rule — at identical path length, and on the lowest state of charge of the flight.

recall in this document is modelled, not measured. There is currently no funded route to measuring it. Of everything listed here this is the omission with the largest effect on a scored outcome, and it is the first item that should be reinstated if the position changes.

The second GNSS receiver is deferred to a measurement, not to funding. It supplies moving-baseline heading, and Section IV-D shows that substituting a magnetometer-class attitude allocation into the case-C budget costs **18 mm**. The honest case for it was never the error budget — it was insurance against a magnetometer sitting near a high-current bus, which the budget does not model. That is an unmeasured risk. **P6 characterises achieved heading accuracy on the built aircraft; if it exceeds roughly 1°, the receiver is reinstated.** If it does not, it was never needed.

The sub-gigahertz safety radio is deferred, and that deferral rests on a configuration rather than on a judgement. The primary link must run in a mode that carries both control and MAVLink telemetry on one link and one autopilot serial port. The alternative transparent-serial mode consumes that link, and an aircraft configured that way has telemetry and no control. **This is verified on the bench at P2; if it does not hold, the radio is not optional.**

Recovery parachutes are permitted, not required. A crash scores -50 against -10 for landing outside the zone, so the deferral is a bet of roughly forty points against the probability of a crash.

Spares cover what crashes, not what fails rarely. Propellers, motors, arms, plate stock and packs are retained. Spare autopilot, GNSS and camera are removed. **This is an accepted risk and I state it as one:** a failure of any of those three inside the flight-test window costs schedule, and the mitigation is procurement lead time rather than inventory.

Flight insurance is deferred and must be confirmed against the rulebook and the Drone Rules before any flight takes place. This is the one deferral that is not purely a performance trade.

B. One caveat on how ground truth will be measured

The ground-truth apparatus is fabricated rather than bought. Survivor targets are clothed forms of the correct size and outline; what a detector responds to is silhouette, scale and contrast against terrain, not anatomy. Two of these items are *better* fabricated: dummy kits can be made to exactly the 200 g that rule C6 fixes, and donated clothing gives wider colour and contrast variety than matched bought sets, which is precisely what detection robustness requires.

The caveat is recorded rather than glossed. The calibration markers are surveyed using the programme's own RTK rover. That is standard practice and adequate for the purpose, but it means the ground truth against which geolocation accuracy is measured is *itself RTK-derived*. The two error sources are therefore correlated, and any measured geotag figure reported in P7–P8 will be slightly optimistic against an independently surveyed datum. Any accuracy claim arising from those phases will state this.

IX. Indigenisation Strategy

Indigenous content is measured, not asserted. Every line in the bill of materials carries a declared Indian value fraction, and the aggregate is computed rather than estimated, which makes the claim auditable.

The configuration adopted for this build carries approximately 36 % Indian content, down from 58 % in the fully specified aircraft. That figure is computed from the per-line Indian fractions of the adopted bill of materials rather than estimated; an earlier draft of this proposal asserted 45 %, which the computation does not support. I report what applies to what is actually being funded rather than the figure that flatters the argument.

Fig. 10 decomposes it, and the decomposition is far more informative than the aggregate. **Propulsion falls from 89 % to 21 % and avionics from 60 % to 32 %** — the two lines

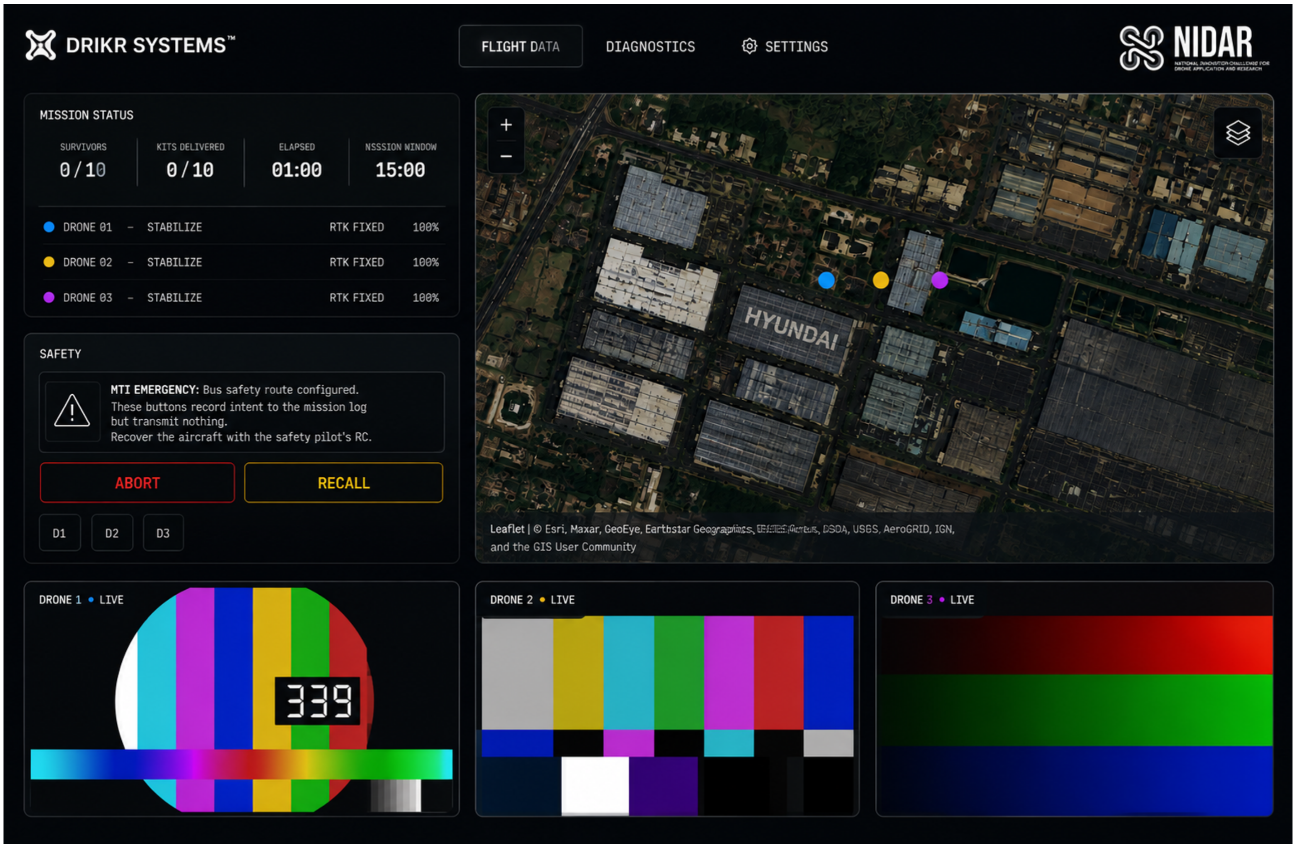

Fig. 9. Ground control station, captured from the running system. A single operator interface carries all eight displays the rules require: mission status, per-aircraft mode, RTK fix state and battery, the shared map, and three concurrent H.264 feeds within the 2.5 Mbps link budget. Map tiles are served from a local cache, so nothing here depends on a network. The safety panel is shown *uncropped*: abort and recall record operator intent in the mission log and transmit nothing, because no downlink command path is yet wired — see Section VII-C. The video panes carry test patterns; no camera has yet flown.

that make the aircraft affordable are precisely the two that make it less Indian. What remains strongly domestic is what the team fabricates or assembles itself: the payload system at 90 % and the airframe structure at 85 %.

This is the clearest consequence of the sourcing decision, and it is reversible. Two lines account for almost all of it: the autopilot and propulsion — motors, speed controllers and certified propellers. Restoring both would return the aircraft to roughly 58 % Indian content, and the Track A and Track B work packages price that reinstatement. Both are drop-in reinstatements requiring no redesign. Component sponsorship from the Indian suppliers surveyed for this bill of materials is the intended route, and it is the reason those suppliers are named throughout rather than described generically.

The inference accelerator is the one genuinely irreducible import, and I declare it plainly rather than obscuring it. Mitigation is confined to what is honest: an Indian carrier board, and Indian enclosure and thermal design. I regard an explicit gap analysis as more credible than a uniformly confident claim, and I note that the domestic RISC-V development kit carried on the aircraft is a genuine demonstration of Indian silicon in flight, albeit in a monitoring role rather than as an accelerator.

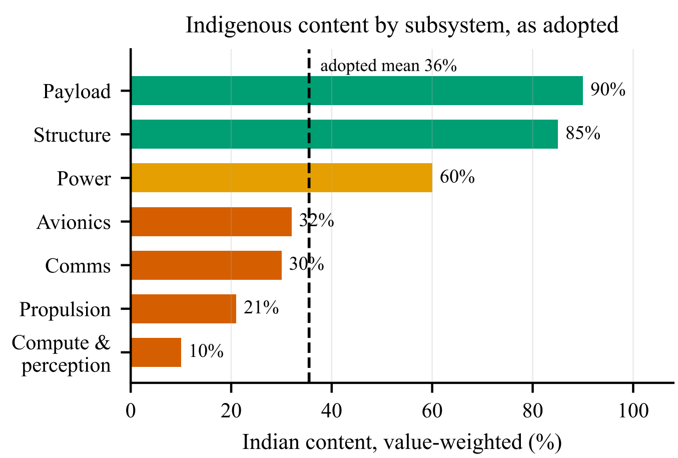

Fig. 10. Indian content by subsystem for the adopted configuration, value-weighted. Four subsystems now sit below 40 %, and between them they account for roughly three quarters of the aircraft by value — which is why the aggregate is worth decomposing rather than quoting alone. What remains strongly domestic is what the team fabricates itself.

X. Risk Register

I draw particular attention to the setup-time risk. It is currently supported by a model with a modest margin and

TABLE IV
INDIGENISATION STATUS BY SUBSYSTEM

Subsystem	Status
Autopilot	*Deferred:* Indian Pixhawk-standard board, domestically made
Propulsion	*Deferred:* Indian motors and ESCs with published thrust data
Airframe	Indian composite fabrication
Cells	Indian, BIS-certified 21700 NMC, 4500 mAh / 45 A. *Retained*
Camera	*Deferred:* Indian embedded-vision supplier
GNSS	Indian receiver, NavIC L1+L5. *Retained*
Co-processor	Indian RISC-V development kit
Inference module	**No Indian equivalent exists at this class**

no measurement, and it gates the entire mission. Timed bench drills are scheduled as the earliest hardware activity in the programme for precisely this reason.

XI. Expected Outcomes and Deliverables

1) Three flight-qualified aircraft with a documented, market-verified bill of materials at approximately 36 % Indian content, and a costed route to restoring the deferred Indian autopilot and propulsion.
2) A single-operator ground control station demonstrated end to end against real aircraft, with no network dependency.
3) A measured detection recall figure against fabricated survivor targets on surveyed ground, replacing the current assumption. Ground truth is RTK-derived rather than independently surveyed, and the resulting figure will state that.
4) A measured delivery circular error probable against a surveyed target, replacing the current model.
5) A measured setup-to-launch time distribution over repeated drills.
6) An open, reproducible evidence base in which every published figure regenerates from its model under continuous integration.
7) A supply-chain report identifying, for each subsystem, the domestic capability that exists, the capability that does not, and what would be required to close the gap.

The seventh deliverable may prove the most durable. The exercise of sourcing a complete airframe domestically, at component granularity and with verified prices, produced a map of Indian UAV supply-chain maturity that I have not seen published elsewhere, including the specific and narrow point at which it currently fails.

XII. Future Work

This proposal deliberately asks for a competition build and nothing more. The work has a longer arc, and I set it out here so that the scope of the present request is unambiguous by contrast.

A. From a competition entry to a deployable system

The entry is not a product. Reaching one requires, in rough order: night and all-weather capability through thermal imaging, which is the single highest-value sensing extension and is excluded here only because it doubles the perception budget; beyond-visual-line-of-sight approval, which is a regulatory programme rather than an engineering one; ingress protection and thermal qualification for monsoon conditions; and endurance beyond the present 15.3 minutes, which at this payload means a different airframe class rather than a better battery.

B. Cost at production volume

The parts basis for this build is single-unit retail for a build of three. The marginal cost of an additional aircraft is already substantially below the per-aircraft figure quoted here, because the ground segment, test equipment and spares are one-off. At production volume the relevant questions are different ones — tooling amortisation, negotiated component pricing, and assembly labour — and none of them can be answered honestly from a three-aircraft bill of materials. I decline to extrapolate them.

C. A route to commercialisation

The natural customers are state disaster management authorities and the national and state disaster response forces. Establishing whether that is a viable business requires three things this programme does not attempt: the size and refresh rate of the addressable fleet, what a comparable capability is currently procured for, and the operating cost per flight hour of the boat or helicopter search it would displace. Those are questions for published tender data and operator interviews, not for a bill of materials, and I would rather name them as open than fill them with plausible figures.

What this programme does produce toward that end is the seventh deliverable in Section XI: a component-level map of where Indian UAV supply chain capability exists and where it does not. That artefact outlives the competition regardless of what follows it.

XIII. Conclusion

I propose an autonomous three-aircraft system that addresses the localisation bottleneck in post-flood search and rescue, sourced domestically wherever that is achievable, and engineered under a verification discipline that distinguishes what has been demonstrated from what has merely been designed. Where an Indian part was deferred it was deferred on cost, not on capability — domestic industry can supply the autopilot and the propulsion, and Section efsec:indig states what restoring them costs.

The system's design point, error budgets and cost basis are established and reproducible. Its coordination and failsafe behaviour have been demonstrated in hardware-in-the-loop simulation across three simultaneous autopilots. What remains is to build the aircraft, fly them, and replace the modelled and assumed quantities in Section VI-A with measured ones.

The resources that requires are set out in the four track work packages, not here. What this document undertakes is narrower and, I think, more useful: that every number in it is checkable, and that the ones which are not yet measured say so.

Acknowledgment

I thank the Department of Electronics and Communication Engineering, Thapar Institute of Engineering and Technology, for institutional support.

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TABLE V
PRINCIPAL RISKS AND MITIGATIONS

Risk	Sev.	Mitigation
Detection recall below target on real imagery	High	**Raised by deferral of the field data campaign.** Public aerial-person datasets carry the model; tiling and fusion parameters remain tunable without hardware change
Inference and software video encode contend on one board	High	The replacement compute has no hardware H.264 encoder. Benchmark both loads together before design freeze; fallback is a reduced tile count on a centre crop
RTK capability unconfirmed on selected receiver	High	Supplier confirmation is a gating procurement action; degraded mode already specified and flagged to the operator
Flight-test window compressed by procurement delay	High	**Raised by deferral of the training airframe and of spares.** Lead time is now the only mitigation, so long-lead items are ordered in tranche 1
Recovery parachute sizing and supply	Medium	No domestic supplier identified; import lead time and correct mass rating are open procurement actions
Setup time exceeds allowance	Medium	Currently modelled, not measured; timed drills are the highest-priority early test
Airframe overhangs the pad edge on touchdown	Medium	Nominal clearance is exactly one airframe radius, so dispersion has no margin. Measure touchdown dispersion in P6 before accepting the slot geometry
Motor thrust below the datasheet claim	High	Selected motors publish no thrust curve, and the stand cannot precede the motors. Thrust is measured in P5 on arrival; a shortfall then forces a propulsion change with the flight-test window already open
Airframe loss during test	High	**Raised by deferral of spares.** Structure set and consumables retained; a second loss in the same week is uncovered
Flight insurance may be mandatory	Medium	Deferred at team direction. To be confirmed against the rulebook and DGCA rules before any flight